# Electromagnetism 2

Lab 1: Joule heat virtual lab

## 1. Objective

## 2. Theory

Part1: Unknown wire measurement (Find $R$ by $V$ and $I$)

$$V = IR$$

Where:

$V$ is voltage [V]

$I$ is electric current [A]

$R$ is wire resistance

Part2: Heating water in isolated system (connect $V$ and measure $T$ with $t$)

$$P_{\mathrm{e}} = \frac{\Delta Q}{\Delta t}$$

$$\frac{V^2}{R} = \frac{m_{\mathrm{w}} c_{\mathrm{w}} \Delta T}{\Delta t}$$

$$\Delta T = \frac{V^2}{R m_{\mathrm{w}} c_{\mathrm{w}}} \Delta t \rightarrow y = mx$$

Then find $c_{\mathrm{water}}$

Where:

$P_{\mathrm{e}}$ is electric power

$m_{\mathrm{w}}$ is water mass

$c_{\mathrm{w}}$ is specific heat capacitor

$\Delta T$ is temperature change

$\Delta t$ is heating time

Part3: Heating ethanol in Isolated system (connect $V$ and measure $T$ with $t$)

$$\Delta T = \frac{V^2}{R\rho_{\text{et}} V c_{\text{et}}} \Delta t \rightarrow y = mx$$

Then find $c_{\text{ethanol}}$

where:

$m_{\text{et}}$ is water mass

$c_{\text{et}}$ is specific heat capacitor

Part4 (Optional): Heating water in closed system (Find cooling constant $k$)

Some of heat may be lost due to the convection. We then assume the heat lost rate by $k\Delta T$, the net power is then

$$P_{\text{net}} = \frac{\Delta Q}{\Delta t}$$

$$P_{\text{e}} - k\Delta T = \frac{m_{\text{w}} c_{\text{w}} \Delta T}{\Delta t}$$

$$\frac{P_e}{m_{\text{w}} c_{\text{w}}} - \frac{k}{m_{\text{w}} c_{\text{w}}} \Delta T = \frac{\Delta T}{\Delta t}$$

$$\frac{\Delta T}{\Delta t} = -\frac{k}{m_{\text{w}} c_{\text{w}}} \Delta T + \frac{P_e}{m_{\text{w}} c_{\text{w}}} \rightarrow y = mx + c$$

Plot $\frac{\Delta T}{\Delta t}$ versus $\Delta T$ to find $k$

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Constant

$c_{\text{w}}$ = 4,184 [J/(kg·K)]

$\rho_{\text{w}}$ = 1,000 [g/cm³]

$c_{\text{et}}$ = 2,440 [J/(kg·K)]

$\rho_{\text{et}}$ = 789 [kg/cm³]

$R$ = 4.000 [Ω]